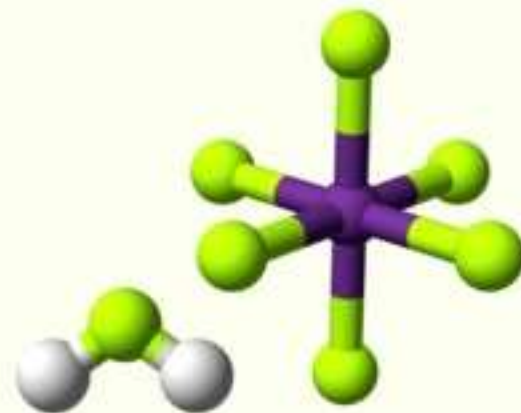


CHEMISTRY

SUBATOMIC PARTICLES

MODULE - 5



SUBATOMIC PARTICLES

ELEMENTARY PARTICLES

Are not composed of any other particles
E.G electrons, bosons etc.

COMPOSITE PARTICLES

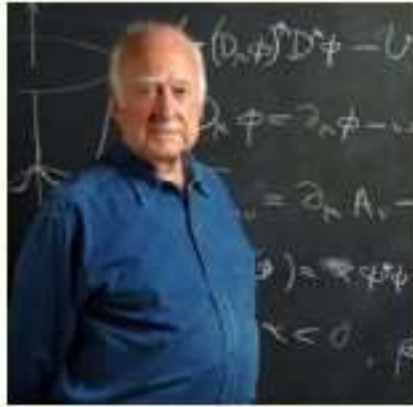
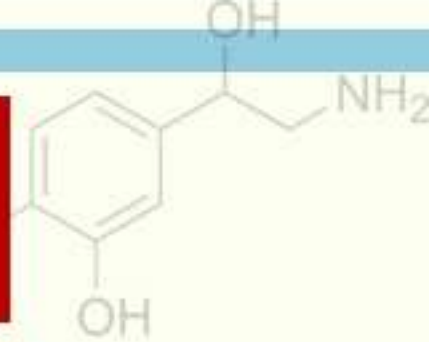
Composed of some elementary particles.
E.G protons, neutrons

What do think why the protons or neutrons are heavier, but electrons are not?



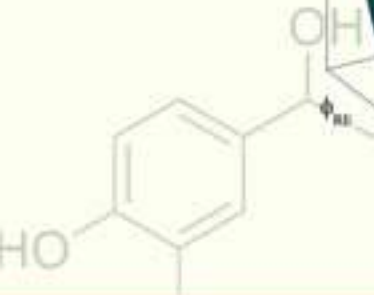
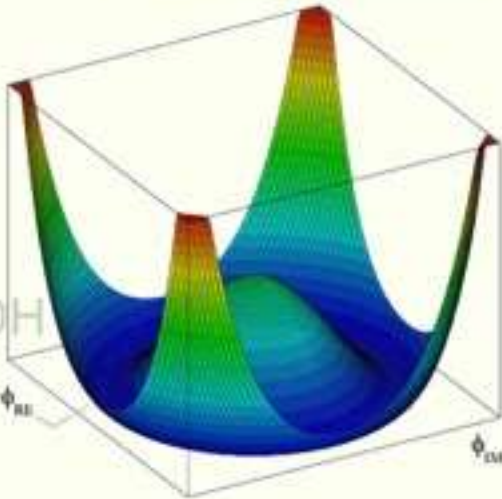


SUB-ATOMIC PARTICLES - WHERE DOES THEIR MASS COMES FROM ?



THE HIGGS BOSON CONCEPT

- Discovered by Peter Higgs in 1964, the Higgs field is an electric field which is the reason why some subatomic particles have a great amount of mass, while some have little and others have none at all!
- This proposed energy field interact with the subatomic particle and give them their mass.
- In the absence of Higgs field, any two subatomic particles would have the same mass or maybe no mass at all!



WHAT IS THE 'BOSON' IN HIGGS FIELD ?



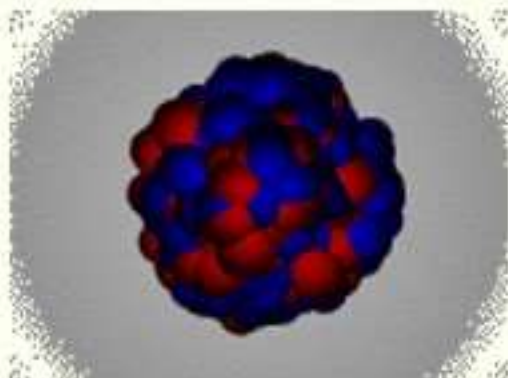
- The smallest bit of the Higgs field is the Higgs Boson.
- It is just the countless amount of tiny particles that make up the Higgs field which interacts with the particles.
- The more Higgs Bosons interacting with the particles the more is the mass, and vice versa.

The Higgs Boson particles were discovered by accelerating subatomic particles in the LHC and slamming them against each other which disturbs the Higgs field.

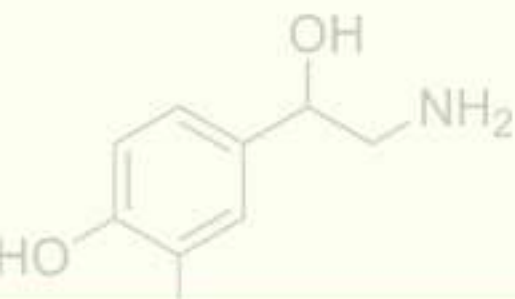
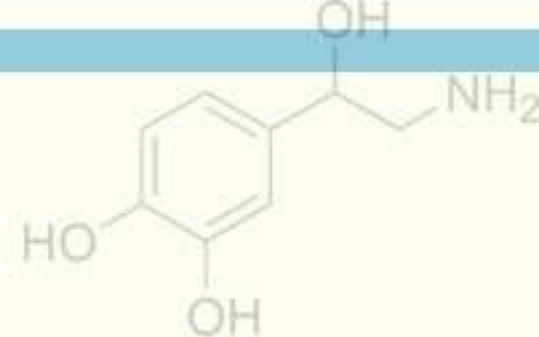


SUBATOMIC PARTICLES

PROTON

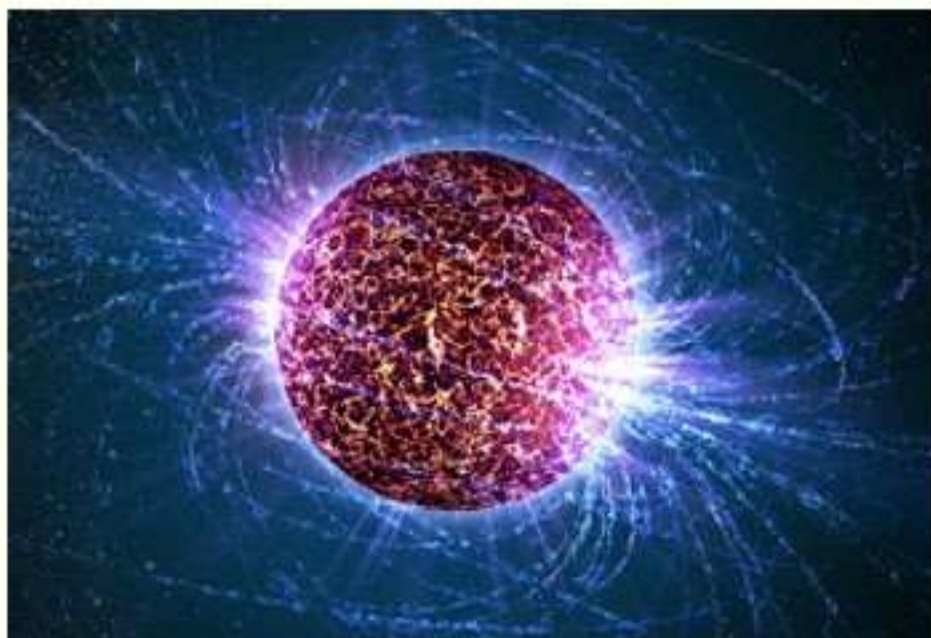


- Proton is positively charged particle inside the atom having +1 charge with a mass about 1u.
- In the Standard Model of Particle physics, protons and neutrons are also called 'hadrons'- a type of composite particles.
- Proton is made up of **three quarks**, which are one of the elementary particles.
- Protons are found in the nucleus of the atom with neutrons.
- Together proton and neutron are also called '*Nucleons*'.

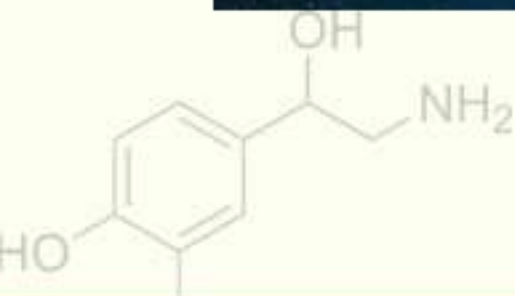
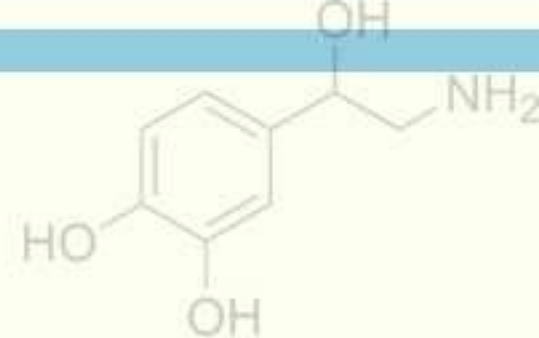


SUBATOMIC PARTICLES

NEUTRON



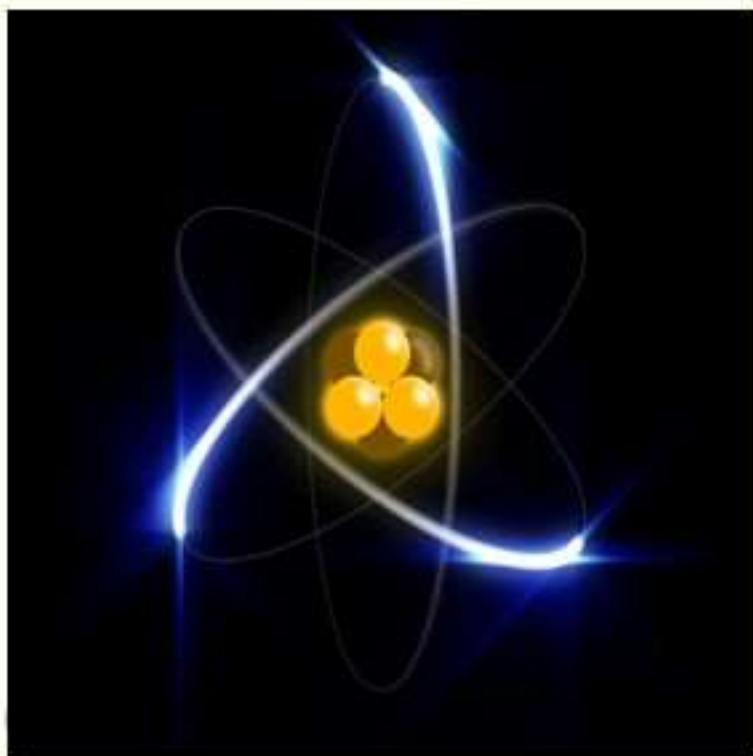
- Neutral part of the atom with slightly higher mass than proton.
- Neutron is also Hadron as it consists of 3 quarks.
- Neutrons are slightly heavier than protons.
- Free neutrons come from cosmic rays are very unstable and reactive.
- Free neutrons can rarely occur in nature and they can be generated to produce Nuclear Energy.



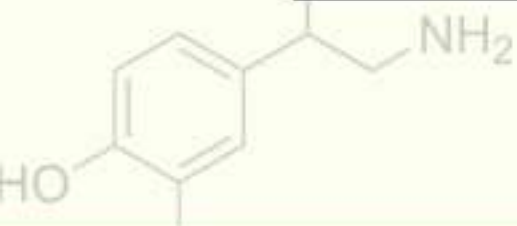
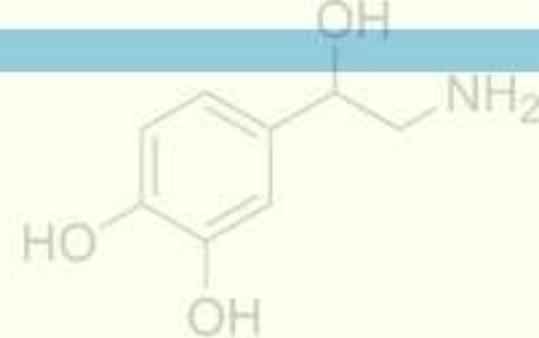


SUBATOMIC PARTICLES

ELECTRON



- The electron is the negatively charged particle in an atom.
- Electrons are extremely lightweight. One proton mass is 1836 times of an electron.
- They continuously revolve around the nucleus.
- Electrons can collide with other particles and can be diffracted like light.
- They play an essential role in electricity, heat conductivity, magnetism etc.





SUBATOMIC PARTICLES FROM THE UNIVERSE

COSMIC RAYS



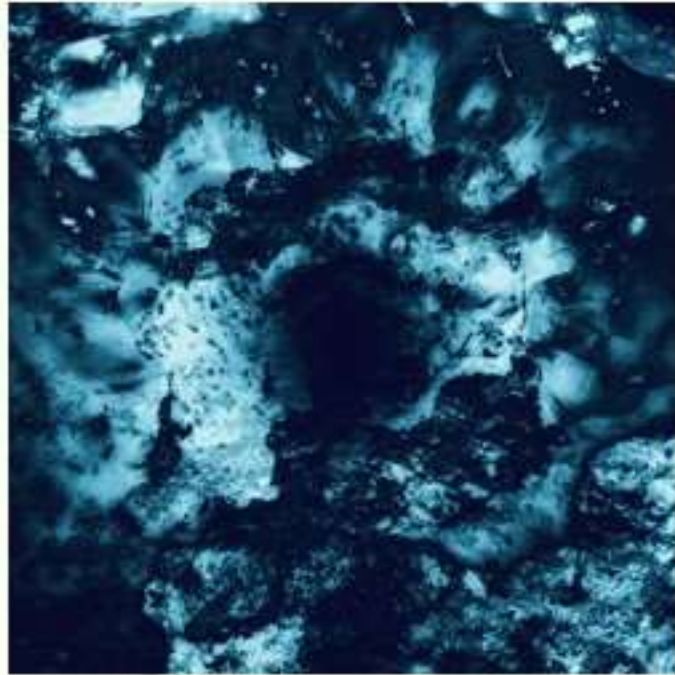
- Many high Energy radiation surrounds our Solar System.
- Due to some Supernova Explosion (Collision of galaxies) these rays are projected towards Earth.
- They are mainly composed of **bare nuclei**.
- A small fraction is also antimatter that is positrons and antiprotons.



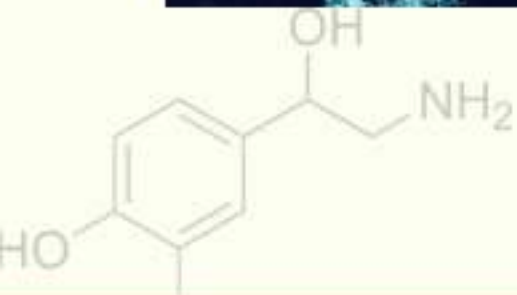
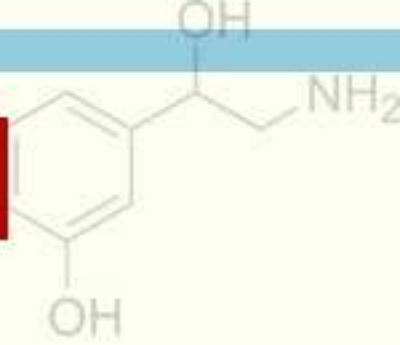


SUBATOMIC PARTICLES FROM THE UNIVERSE

ANTIMATTER

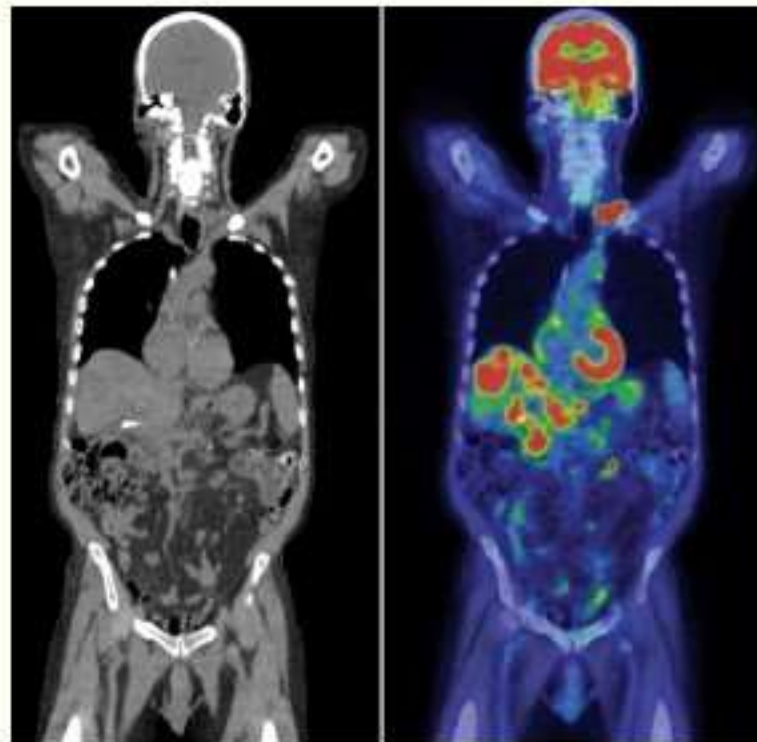
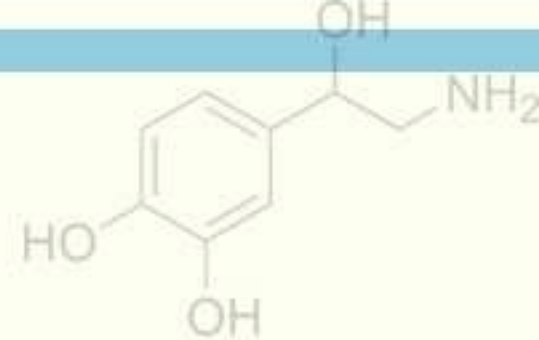


- Composed of antiparticle partners of regular matter.
- They compose of Positron (the antiparticle of the electron) & antiproton (the antiparticle of the proton).
- They are difficult to produce and store because they get destroyed on coming in contact with any matter.
- Antimatter is naturally produced in the Universe where high energy collisions take place such as cosmic rays.
- Institutions like CERN- European Centre for Nuclear Research etc. Have prepared antimatter and are trying to utilize it for mankind.





USES OF ANTIMATTER



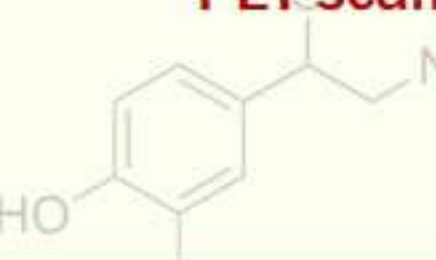
**PET scans of vital Human
Organs**

PRESENT APPLICATIONS

- Medical Imaging is one practical application of Antimatter in present day such as **Positron Emission Tomography(PET)**.
- Antiproton is said to treat Cancer.

FUTURE APPLICATIONS

- Antimatter can be used in future as a fuel for interstellar travel.
- Can be used as ammunition for nuclear wars.





HOW ARE ELECTRONS DISTRIBUTED IN AN ATOM?

1

THE $2n^2$ RULE



K = 1 first shell

According to $2n^2$ Rule

$$K = 2 \times (1)^2$$

$$= 2 \times 1 \times 1 = 2 e^-$$

Similarly,

L = 2 second shell

$$L = 2 \times 2 \times 2 = 8 e^-$$

$$M = 2 \times 3 \times 3 = 18 e^-$$

$$N = 2 \times 4 \times 4 = 32 e^-$$

Any atom can accommodate this fixed number of e^- in their respective shells

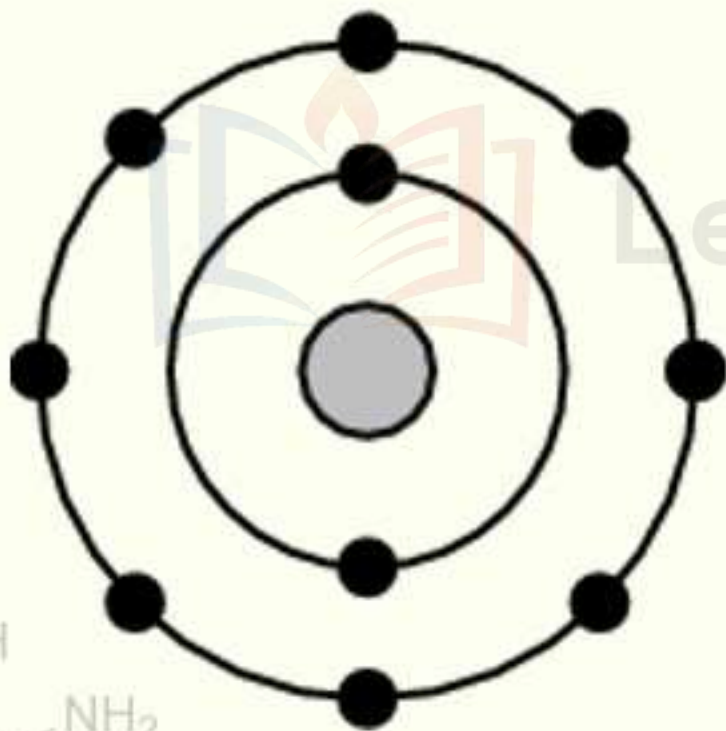




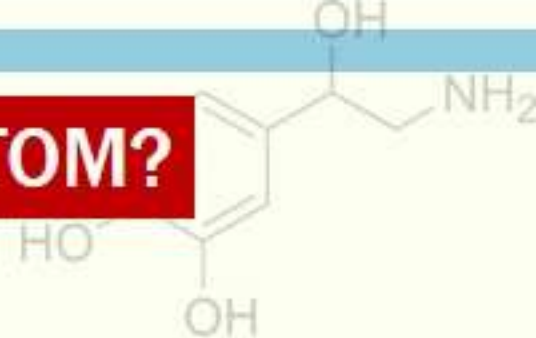
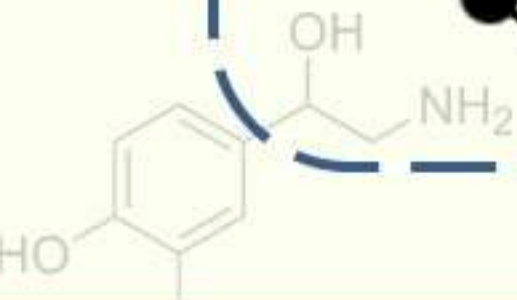
HOW ARE ELECTRONS DISTRIBUTED IN AN ATOM?

2

OCTET RULE

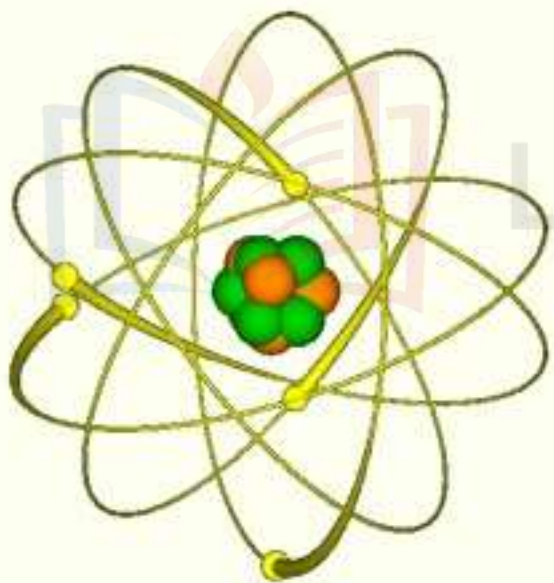


- The maximum number of e^- accommodated in the last shell is 8 (Bohr-Bury Scheme)
- In case of Hydrogen and Helium they have Duplet Rule, they should have 2 electrons in the outermost shell.

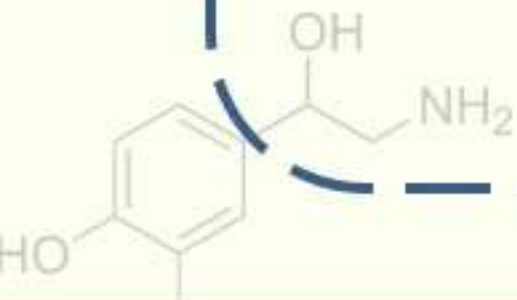
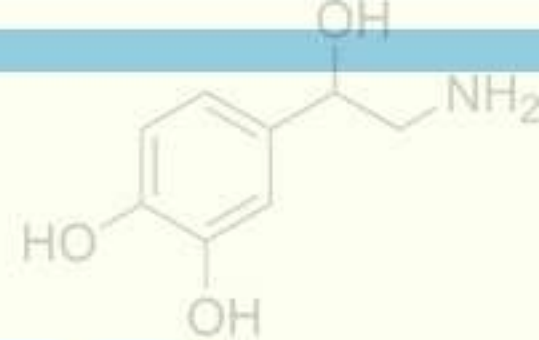




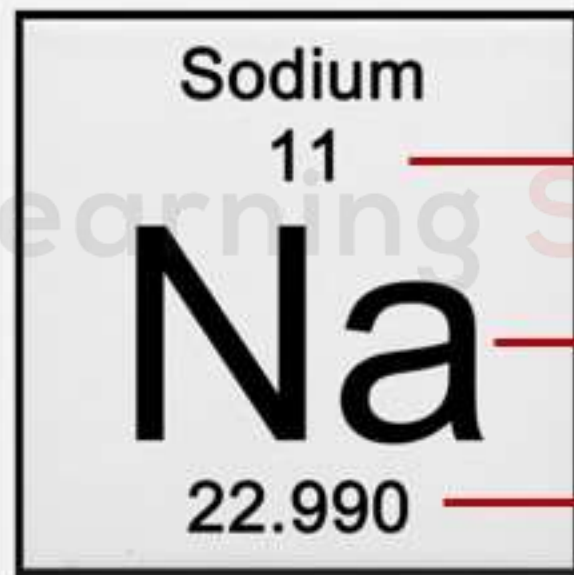
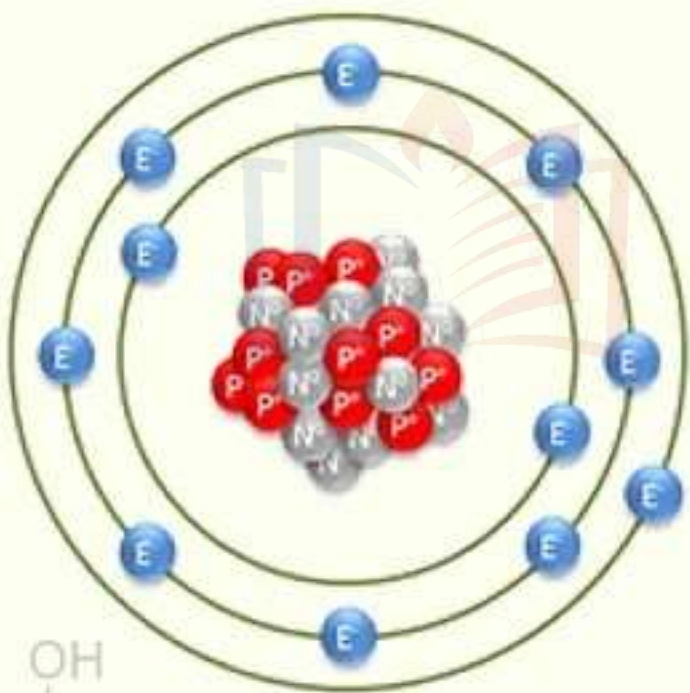
VALENCY OF AN ATOM



- Valency is the combining capacity of an atom.
- An atom can either donate or accept electrons in order to attain a stable configuration.
- Metals have positive valency while non-metals have negative valency.



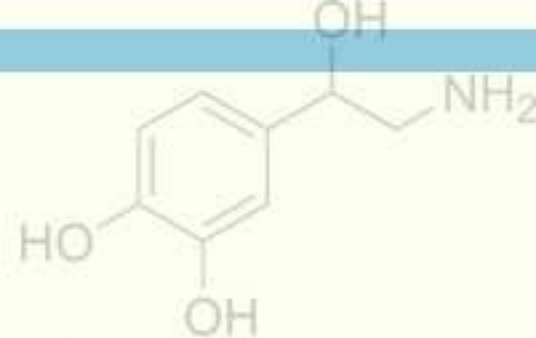
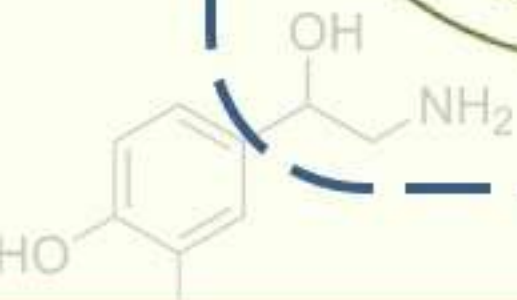
ATOM – HOW IS IT REPRESENTED?



→ ATOMIC NUMBER

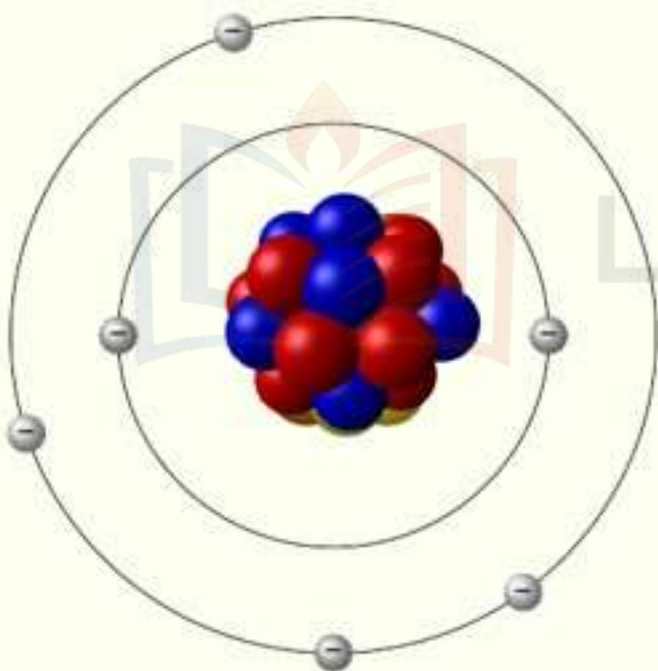
→ SYMBOL

→ ATOMIC MASS

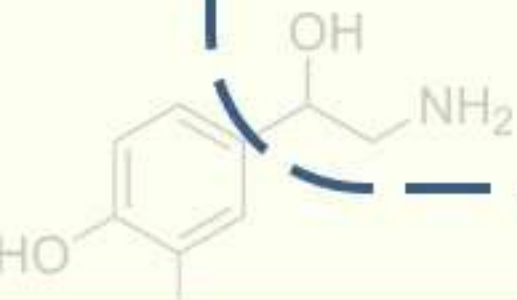
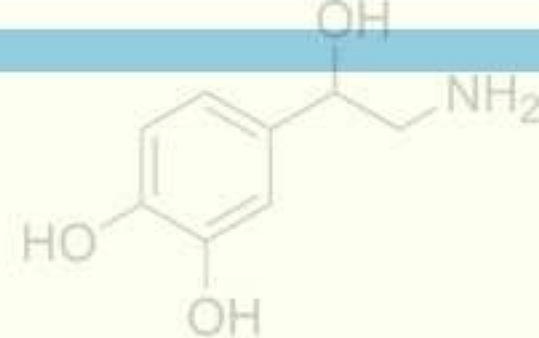




ATOMIC NUMBER

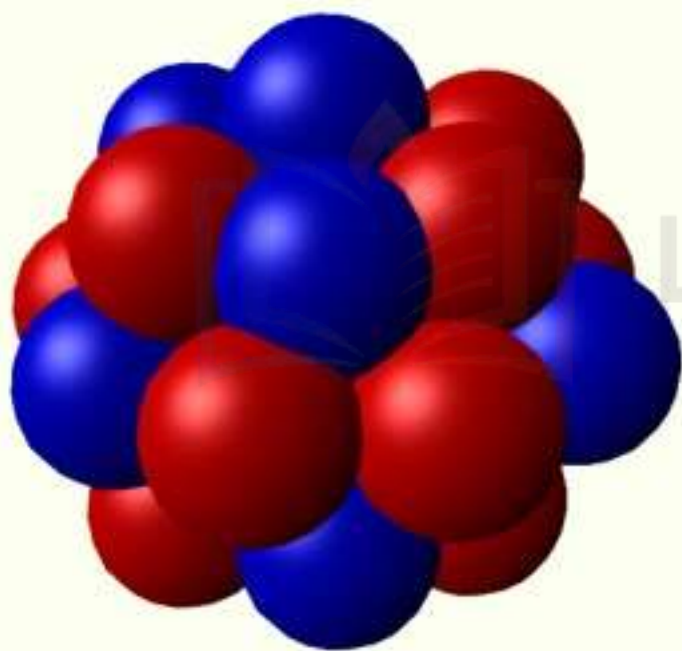


- Atomic number is the number of protons or the number of electrons in the atom.
- An atom is electrically neutral, that means the no. of protons is always equal to the no. of electrons.
- Every atom has a unique atomic number.
- It is represented by the letter Z.





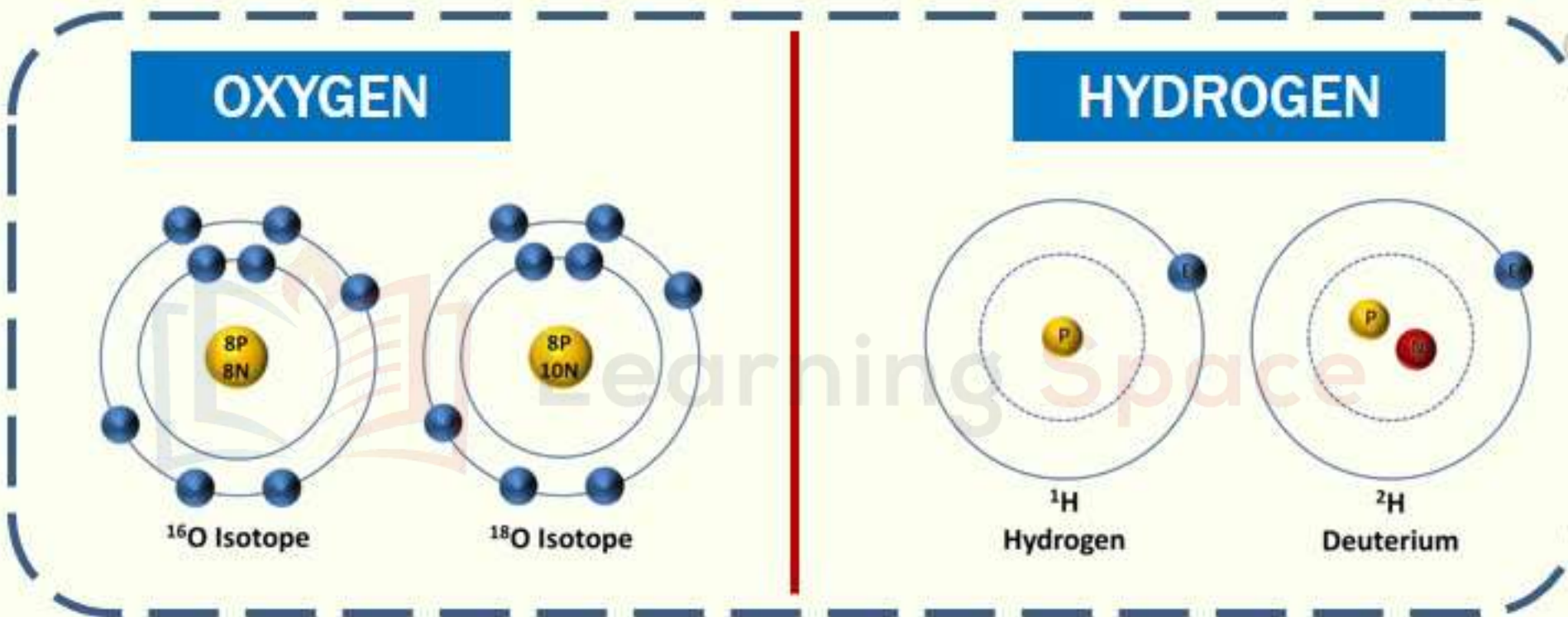
ATOMIC MASS



- The mass of the nucleons is the atomic mass and represented by letter A.
- It is simply the number of protons and neutrons as their mass is approx. 1u.
- It is measured in atomic mass units a.m.u or simply u.
- The atomic number can be found out from the atomic mass by
$$Z = A - \text{no. of neutrons}$$



SAME ELEMENTS – DIFFERENT ATOMIC MASSES!

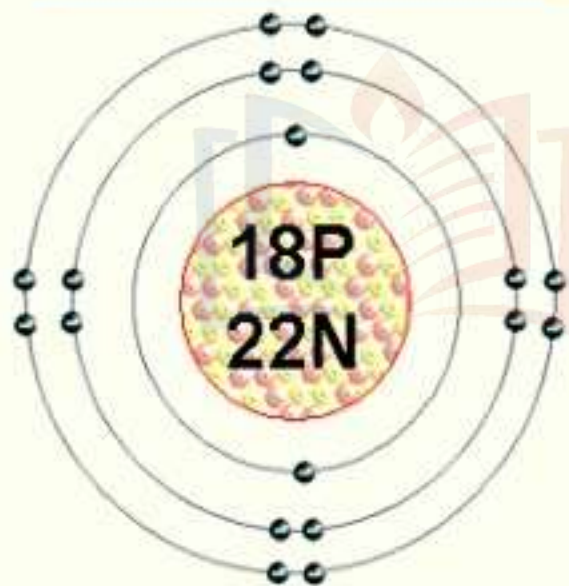


- The atoms which have same atomic number but different atomic mass are called isotopes.
- There are 90 naturally occurring elements with roughly 250 stable isotopes, and over 3200 unstable or radioactive isotopes.



DIFFERENT ELEMENTS – SAME ATOMIC MASSES !

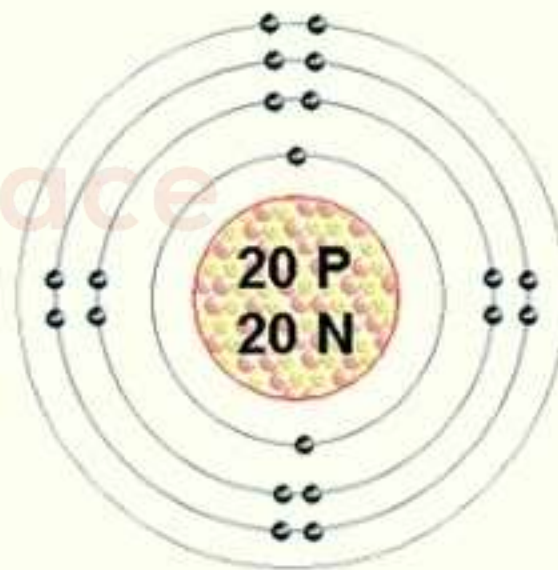
ARGON



POTASSIUM



CALCIUM



ISOBARS – the atoms which have same atomic mass but different atomic numbers.





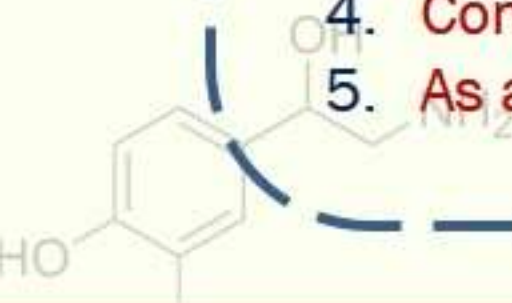
ISOTOPES

THE VARIETY OF APPLICATIONS



Different isotopes of the same element often have completely different properties – making some of them invaluable for mankind

1. **Radiopharmaceuticals used for medical imaging** in the diagnosis of a wide range of ailments, from pneumonia to heart problems to cancer;
2. **Radiopharmaceuticals for cancer treatment** and other therapeutic applications;
3. **Batteries that power NASA satellites** in the far reaches of our solar system;
4. **Control rods** that prevent nuclear power reactors from melting down;
5. **As a "fingerprint"** used in forensic analysis.



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